Particle CLEM in PeleC

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LEM and LES

LES (PeleC, MOL–RK2), Y_k not solved here

$$U_c = \big[\rho,\ \rho\mathbf{u},\ \rho E\big], \quad E = e + \tfrac{1}{2}|\mathbf{u}|^2$$

Line $\mathcal{P}_c$ AMReX particles in *every* cell of *every* level, ordered by `lem_index` (head $= 0$)

$$p \in \mathcal{P}_c: \ \big(m_p,\, v_p,\, \rho_p,\, T_p,\, e_p,\, P_p,\, Y_{k,p}\big),\ \ \rho_p = \frac{m_p}{v_p}$$

Conserved filter

$$[\rho\phi]_c = \frac{1}{V_c}\sum_{p\in\mathcal{P}_c} m_p\,\phi_p, \qquad \phi \in \{1,\ Y_k,\ e\}$$

Mass-weighted filter

$$[\phi]_c = \frac{1}{m_c}\sum_{p\in\mathcal{P}_c} m_p\,\phi_p, \qquad \phi \in \{1,\ Y_k,\ e\}$$

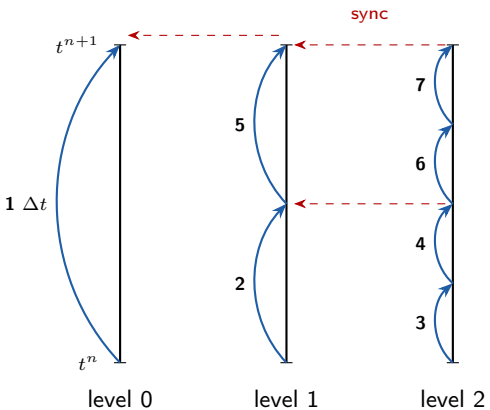
Ownership

- LES: ρ, $\rho\mathbf{u}$, ρE
- lines: Y_k and e(needed for Reaction) (transport, diffusion, reaction)

Face mass flux kept per level, RK2 average

$$F = \rho^* u^* A, \quad \mu_f = \Delta t\,|\dot{m}_f|$$

PeleC AMR time stepping

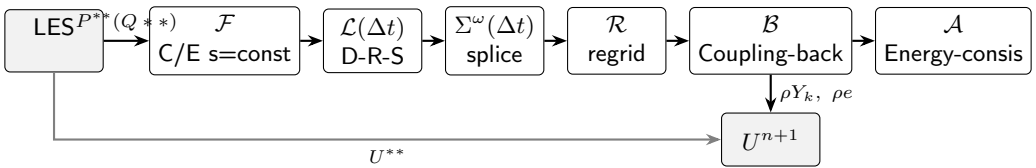


Subcycling in time, refinement ratio $r = 2$

$$\Delta t_\ell = \frac{\Delta t_0}{r^\ell}$$

- Reflux is the syncronization of the fluxes at the coarse/fine interface. Not trivial in particle advection.
- Need to make decision of what information can be lost,

One step as a composition of operators



Level step (ℓ, its own Δt_ℓ, in place of `react_state`)

$$\mathcal{P}_\ell^{n+1} = \mathcal{A} \circ \mathcal{B} \circ \mathcal{R} \circ \Sigma^{\omega}(\Delta t_\ell) \circ \mathcal{L}(\Delta t_\ell) \circ \mathcal{F} \left[\mathcal{P}_\ell^{n}\right]$$

Coarse/fine sync (`post_timestep` of ℓ) and **AMR regrid**

$$\mathcal{P}_\ell \leftarrow \mathcal{D}_{\ell+1\to\ell} \circ \mathcal{B}_\delta \circ \mathcal{R}\big|_\partial \circ \mathcal{M}\left[\mathcal{P}_\ell\right], \qquad \mathcal{P} \leftarrow \mathcal{D} \circ \mathcal{K} \circ \mathcal{C}\left[\mathcal{P}\right]$$

Particle reflux, regrid, couple-back, and average down. δm is conserved(not necessarily $\delta\phi$)

Line operations

$\mathcal{F}$ **forward coupling** (isentropic, frozen Y),
$\Pi = P^{**}/P_p$
$\mathcal{L}$ **diffusion** (mass coordinate $dm = \rho A\, dx$, isobaric, periodic line)

$$\frac{DY_k}{Dt} = RHS(Y_k)$$

$$\frac{DT}{Dt} = RHS(T) + \frac{1}{\rho c_v} \sum_k h_k\, RHS(Y_k)$$

explicit integration

Σ^ω **splice**, outflux faces N_c^+, influx faces N_c^-

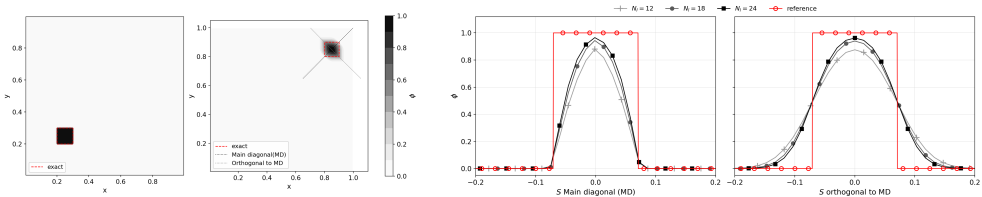
$$\sigma_f = \mathrm{Cut}_{\mu_f}\big(\text{head of } \mathcal{P}_c\big), \quad f \in \mathcal{O}^\omega(N_c^+)$$

$\mathcal{O}^+$: $|\mu|$ descending, $\mathcal{O}^-$: ascending, alternating: $\omega_n = (-1)^{n+1}$
Strang composition $\mathcal{O}^+ ... \mathcal{O}^-$

$\mathcal{R}$ **regrid** to n_{lem} equal-mass elements. Conservation of m, v, $\sum mY_k$, $\sum me$

Particle advection: splicing

Advection of a square-shaped scalar field along the diagonal (2D, single level)



scalar along the main diagonal (left) and orthogonal to it (right) for $N_l = 12,\ 18,\ 24$ particles per cell

- a three-dimensional transport represented by one-dimensional lines; the total scalar is conserved by the particle method
- the numerical diffusion is orthogonal to the mean flow direction, and decreases with the number of particles

Coupling back $\mathcal{B}$: LEM lines $\rightarrow$ LES

Species: conserved variation of a filtered scalar

$$\Delta[\rho Y_k] = [\rho Y_k]_c^{\text{post}} - [\rho Y_k]_c^{\text{pre}}, \qquad \widetilde{\rho Y_k} = (\rho Y_k)^n + \Delta[\rho Y_k]$$

mass species projection

$$(\rho Y_k)^{n+1} = \rho^{**} \frac{\widetilde{\rho Y_k}}{\sum_j \widetilde{\rho Y_j}} \qquad \Longrightarrow \qquad \sum_k Y_k^{n+1} = 1$$

Total energy: the LES accounts for the subgrid effect

$$(\rho E)^{n+1} = (\rho E)^{**} + \Delta[\rho e]_{\mathcal{L}}, \qquad \Delta[\rho e]_{\mathcal{L}} = [\rho e]_c\big|_{\text{after } \mathcal{L}} - [\rho e]_c\big|_{\text{before } \mathcal{L}}$$

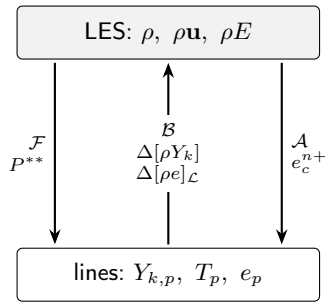
Internal energy consistency $\mathcal{A}$: LES $\rightarrow$ lines

$$\delta e_c = e_c^{n+1} - \bar{e}_c, \qquad \bar{e}_c = \frac{\sum_p m_p e_p}{\sum_p m_p}$$

δe_c internal energy difference between LES and LEM(relevant in AMR)

$$e_p \leftarrow e_p + \delta e_c$$

Shift from particle internal values to make internal energy consistency.



at t^{n+1} LES and lines agree in mass, energy and composition; the LES stays conservative

Species and their enthalpy flux effect

Problem: the species are frozen during RK_2, energy and species diffusion at LES are set zero $\mathbf{J}_k$ (consistency between species diffusion and energy transport)

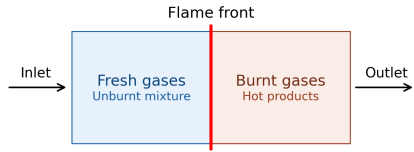
$$\partial_t(\rho Y_k) + \underbrace{\nabla\cdot(\rho Y_k\mathbf{u}) + \nabla\cdot\mathbf{J}_k}_{=0 \text{ (frozen)}} = 0, \qquad \partial_t(\rho E) + \cdots = \cdots - \nabla\cdot\left(\textstyle\sum_k h_k\mathbf{J}_k\right)$$

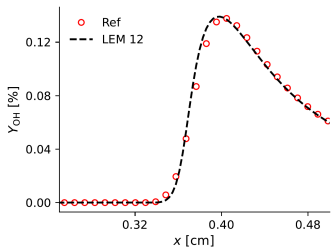
Fix: when the lines own the species, both are dropped on the same face

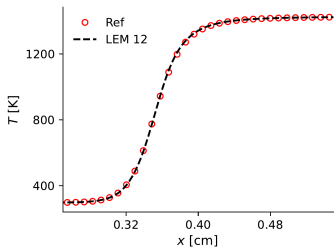
$$\mathbf{J}_k = 0 \quad \text{and} \quad \sum_k h_k\mathbf{J}_k = 0 \qquad \text{in the LES}$$

Premixed flame

Lean H_2–air premixed flame, $\phi = 0.4$, 300 K, 1 atm: 128 cells $\times$ 12 particles against a 512-cell reference

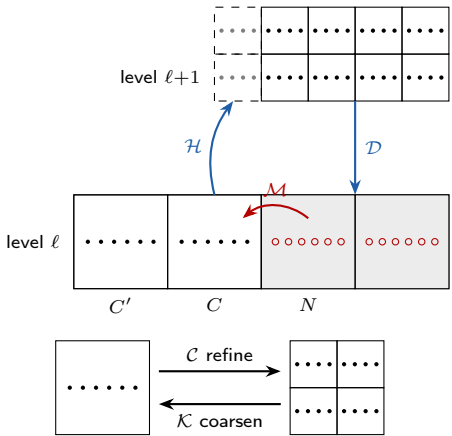






OH mass fraction (left) and temperature (right); the heat release is carried by the particle internal energy, the small shift in the species curve comes from the splicing

AMR: lines on every level



$\mathcal{H}$ **halo**: clone of the coarse line, same state, $(m_k, v_k)/r^d$

$\mathcal{M}$ **particle mass reflux**, per interface face

$$\delta\mu_f = \sum_s \mu^s_{f,\text{fine}} - \mu_{f,\text{coarse}}$$

Reflux follows the splicing rule. Then regriding $\mathcal{R}$ on C, and species coupling following coupling-back.

$\mathcal{D}$ **average down**, element k over the r^d children

$$m_k = \sum_c m_{c,k}, \quad v_k = \sum_c v_{c,k}, \quad \psi_k = \frac{\sum_c m_{c,k}\,\psi_{c,k}}{m_k}$$

$\psi \in \{e,\ Y_k\}$, then $T_k = T(\rho_k, e_k, Y_k)$

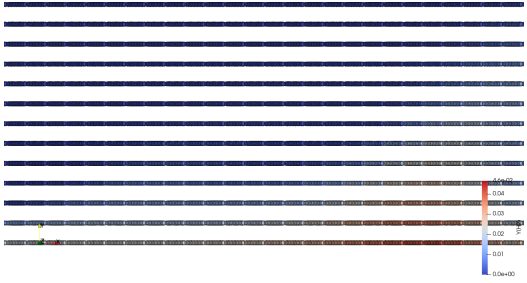
$\mathcal{C}$ **refine**: children are clones with m/r^d; LES filled piecewise constant

$\mathcal{K}$ **coarsen**: fine lines dropped; the parent already holds $\mathcal{D}$

Particle topology on the AMR hierarchy

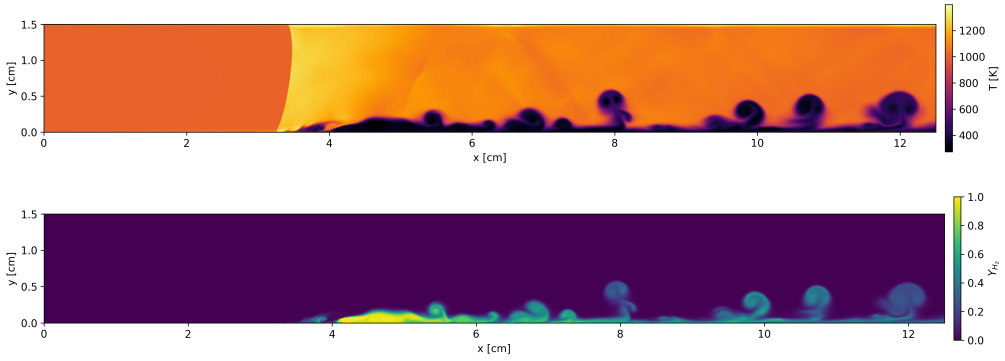


one LEM line per cell on every level: the lines follow the block structure and are denser on the finer levels



elements coloured by Y_{H_2}: the sub-grid structure carried along each line

H_2 jet in supersonic crossflow (2D), $t = 0.25$ ms



temperature (top) and H_2 mass fraction (bottom)